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ELECTRICAL CONDUCTIVITY IN STRUCTURALLY INHOMOGENEOUS DISCONTINUOUS METAL FILMS*

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1. INTRODUCTION

Conduction in discontinuous films has been studied by many authors[1]. However, papers devoted to the relation between the electrical conductivity and film structure are scarce. The problem was given much consideration in ref. 2, where the author investigated the conduction mechanism in films of four structural types. All four types were those found in structurally homogeneous films however, *i.e.* films with a statistically homogeneous distribution of islands and their dimensions over the whole film area. The purpose of the present paper is to investigate the effect of structural inhomogeneity in discontinuous films.

2. EXPERIMENTAL AND RESULTS

The study of film electrical conductivity was accompanied by a structural investigation using the electron microscope, and measurements of the potential distribution were taken by scanning and emission electron microscopy. Structurally homogeneous films (Fig. 1(a)) can be obtained by the thermal evaporation method on an amorphous

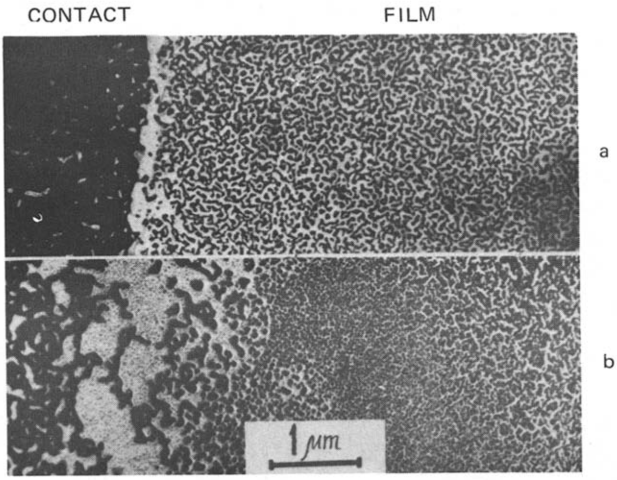

Fig. 1. Structure of (a) homogeneous and (b) inhomogeneous discontinuous gold films near the contact

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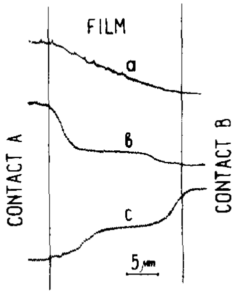

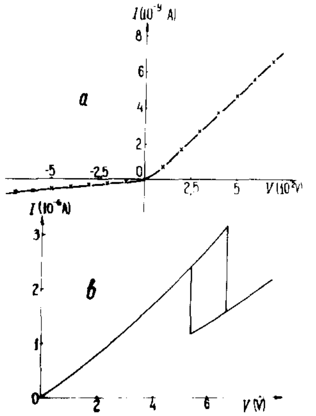

Fig. 2. Distribution of potential in discontinuous gold films. A, B, contacts; a, homogeneous film; b and c, inhomogeneous film; b, contact A positive; c, contact B positive.

Fig. 3. Current–voltage characteristics of structurally inhomogeneous gold films: (a) film with rectifying effect; (b) film with switching effect.

dielectric substrate between two sharply delineated contacts. Such contacts are obtained by scratching a streak on a continuous metal film or by photolithography.

In structurally homogeneous films the potential is on average distributed uniformly from contact to contact, as shown in Fig. 2(a). The conduction current in the low voltage region obeys Ohm's law, but with further increase in voltage V the current increases rapidly to values where an irreversible rearrangement of the film structure is induced, owing to an excessive power output. In this voltage range the current–voltage characteristics of the conduction current I do not depend on the polarity of the voltage applied to the film, and the voltage dependence of the film resistance $R_V = V/I$ measured in the pulse regime is $R_V = R_{V(0)} - \text{const. } V$. This dependence is distorted by heating in the direct current regime.

The electrodes were mostly prepared by thermal evaporation using the mask method. In this case contacts with sharp borders were never obtained since the material of the electrodes penetrated beneath the mask edges. As early as the stage of electrode preparation, inhomogeneously structured island films were formed in the regions near the electrodes. Inhomogeneities near contacts also remained after evaporation of the film under investigation (Fig. 1(b)). When the electrode material was equally distributed under both edges of the mask symmetrically inhomogeneous films could finally be obtained. In such films the main potential drop occurred in the near-contact regions (Figs. 2(b), (c)). More unexpectedly, the potential distribution was found to depend on the polarity of the voltage applied to the film: the potential drop was always large at the positive electrode. In films with inhomogeneous structure of this type therefore the electric field, resistance and current output power in the vicinity of the positive electrode were always larger than those averaged over the film. It is clear that in this case the positive electrode was heated to a higher temperature than the negative, as confirmed by the experiment. Thus, in films with inhomogeneous structure a temperature gradient is present along the film; it depends on the thermal conductivity and geometry of the film and substrate.

Films with irregular structural inhomogeneity result from evaporation onto cleaved surfaces of NaCl, KBr and other crystals. This is due to defects in the form of steps etc. remaining on the cleaved surfaces.

Films can be obtained with controlled asymmetric structure with respect to the electrodes. This implies that a certain structure near one electrode transforms gradually to another structure near the other electrode. Such films demonstrate a dependence of their conductivity on the polarity of the applied voltage, *i.e.* a rectifying effect (Fig. 3(a)). The characteristic obtained is stable and reproducible.

The phenomenon of n-type switching observed both in a system of several discontinuous films connected in parallel[3] and in a single film[4] has been reported in the literature. In the latter case, however, the films with a switching effect were obtained casually and after several switchings no effect was observed. We have found the switching to be due to the presence of structural inhomogeneities in the films. Using specially treated gold films, we obtained stable switching elements with the characteristic shown in Fig. 3(b). Hundreds of switching cycles did not lead to any changes in the characteristics.

3. DISCUSSION

The distinctive feature of the current-voltage characteristic for both homogeneous and inhomogeneous films is their deviation from ohmic behaviour towards a faster increase in current. This result is suggested as being due to the heating of the electron gas in the islands[5] arising from the specific structure of discontinuous films. Since the island dimensions are smaller than the mean free path of electrons in the metal the scattering by phonons decreases, leading to a higher temperature of the electron gas in the islands compared with that of the island lattice.

For an understanding of the conduction current characteristics in structurally inhomogeneous films (dependence of the potential distribution on the polarity, rectifying and switching effects) important factors, in our opinion, are the charges in the islands appearing when current passes through the film[6]. Reference 7 presents in general form a solution of the problem on charge distribution in the island system with passage of current. It is demonstrated that in structurally inhomogeneous films, owing to the relation between charges in the islands, their potentials and the tunnelling probability between islands, the potential drop in the film and the current through it must be sensitive to the manner of connecting the voltage source to the contacts, *i.e.* to their grounding or to the polarity of the voltage applied. It may also be that in these films a space charge region is formed, as a result of which a sag in the potential distribution curve is to be expected. The increase in space charge with further increase in current will deepen the potential minimum, involving a decrease in the electron drift velocity and resulting in a new increase in the space charge; an instability in the n-type conduction current will develop.

Whether the theory[7] describes the systems under study or not, our investigations of films with inhomogeneous structure have shown that they offer new physical challenges and suggest previously unforeseen areas for practical application.

CONCLUSION

Our investigations show that the structure of island films essentially influences the character of electrical conductivity and determines the existence of stable rectification and switching effects. It is considered likely that structurally inhomogeneous films offer other possibilities. A study of the mechanisms of the effects observed and the preparation of films with a programmed structure would seem to be indicated.

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